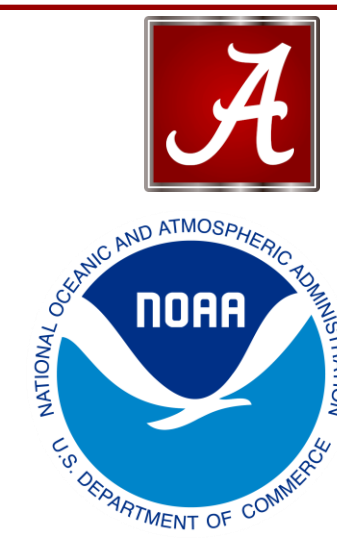




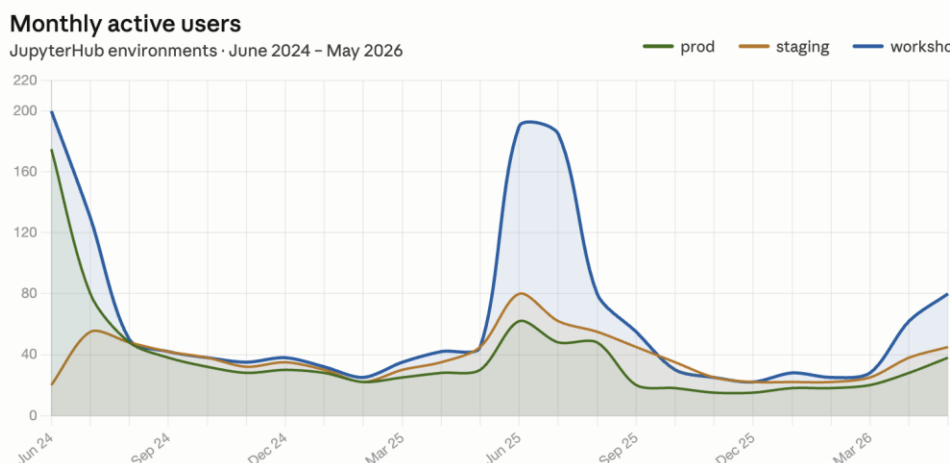
Goals

The CIROH cyberinfrastructure team develops and supports the cyberinfrastructure essential for community water resource modeling and NOAA and USGS projects. This includes building robust, flexible, and scalable platforms that enable researchers to efficiently deploy and execute complex hydrological simulations while handling diverse data sources and computational demands.

Cyberinfrastructure (Cloud and On-premise)

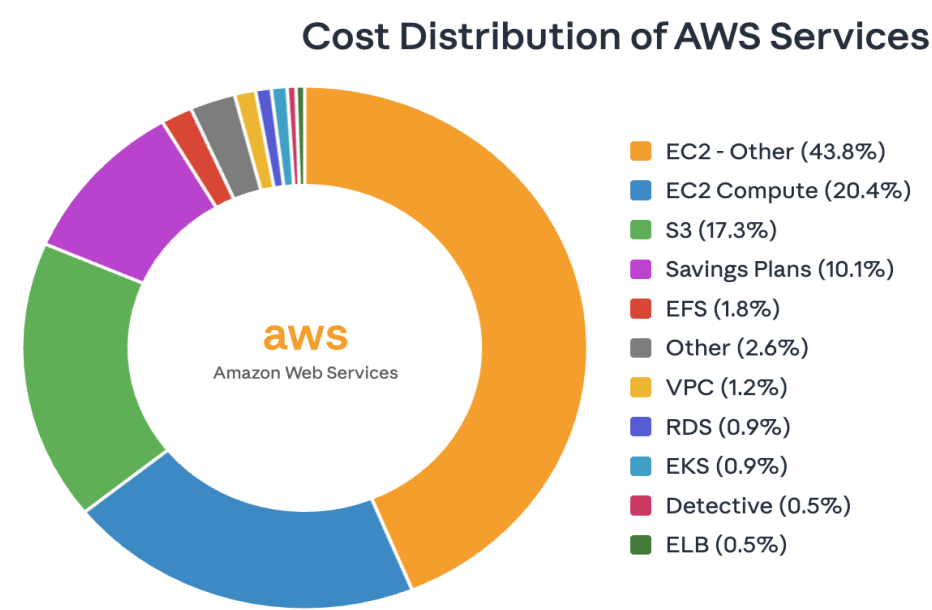
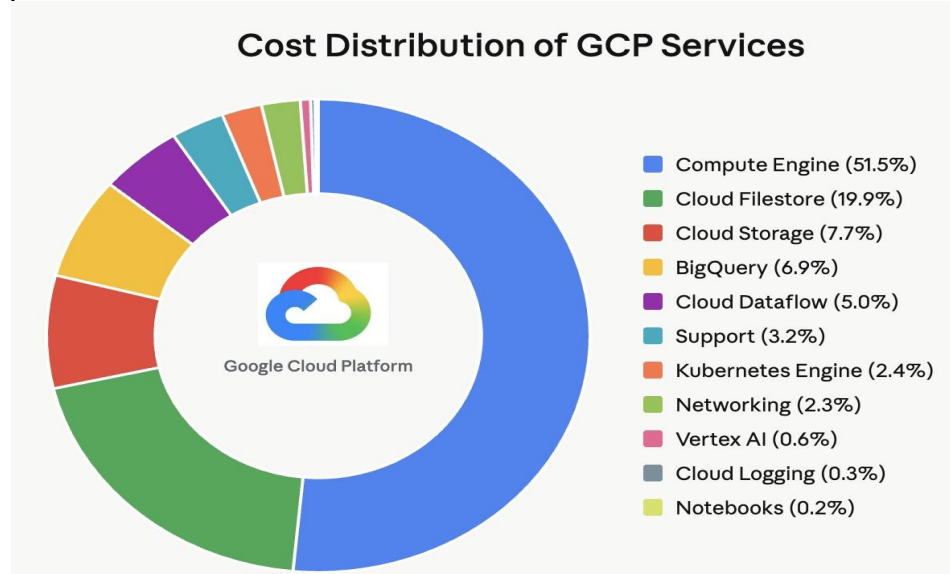
1. Seamless Research Collaboration

- Strengthened CI/CD pipeline within the 2i2c JupyterHub platform on GCP, enabling efficient continuous development for collaborative projects.
- Integrated 2i2c with CUAHSI HydroShare.



2. Cloud and HPC Resource Management

- Supports 100+ CIROH projects and 500+ researchers.
- Implemented cost monitoring and alerts for proactive notifications.



3. Advance Computing for Machine Learning and Hydrological Research

- 2i2c – GPU cluster
- Pantarhei, Wukong and NSF ACCESS allocations

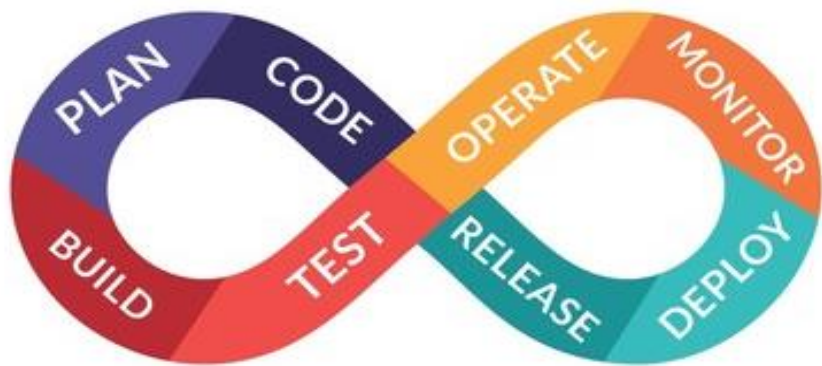
4. Cyberinfrastructure for Community and Events

- Support major events likes DevCon, Summer Institute and Conferences workshops.

5. Data Access and Management

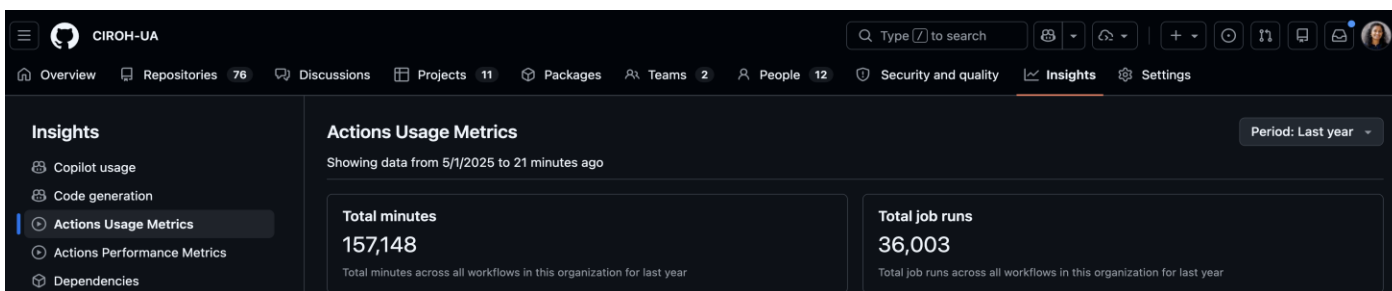
- Deployed and maintain the NWM BigQuery API on GCP.

DevSecOps Practices (CIROH-UA GitHub)



Build & release

- Automated CI pipelines validate every change with tests, security scans, and policy checks
- Defects, vulnerabilities, and misconfigurations are caught before code reaches shared environments



Security controls

- Continuous scanning of containers, code, and infrastructure for vulnerabilities, secrets, and misconfigurations
- Least-privilege IAM and OIDC-based authentication eliminate long-lived credentials across the cloud footprint

Release & Deploy

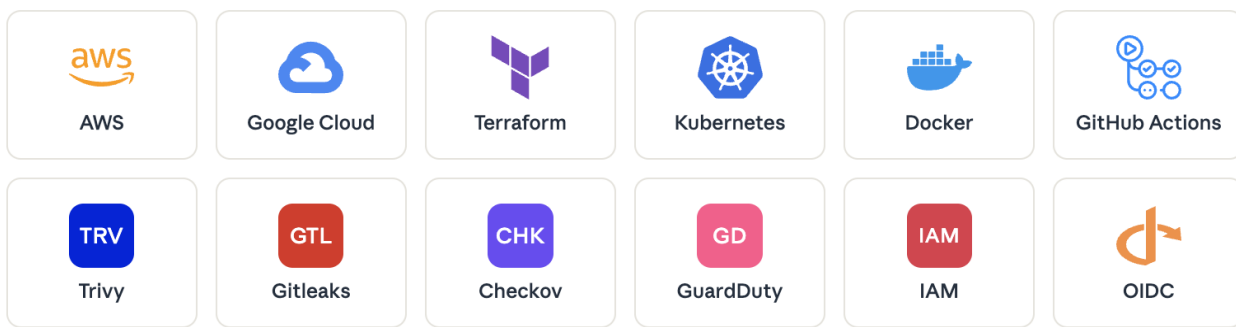
- Declarative configurations and versioned artifacts make deployments repeatable, auditable, and reversible
- Promotion through environments is gated and reproducible

Operate & Monitor

- Live services across distributed compute environments are continuously observed for health, performance, and cost
- Alerts and dashboards enable proactive response to anomalies

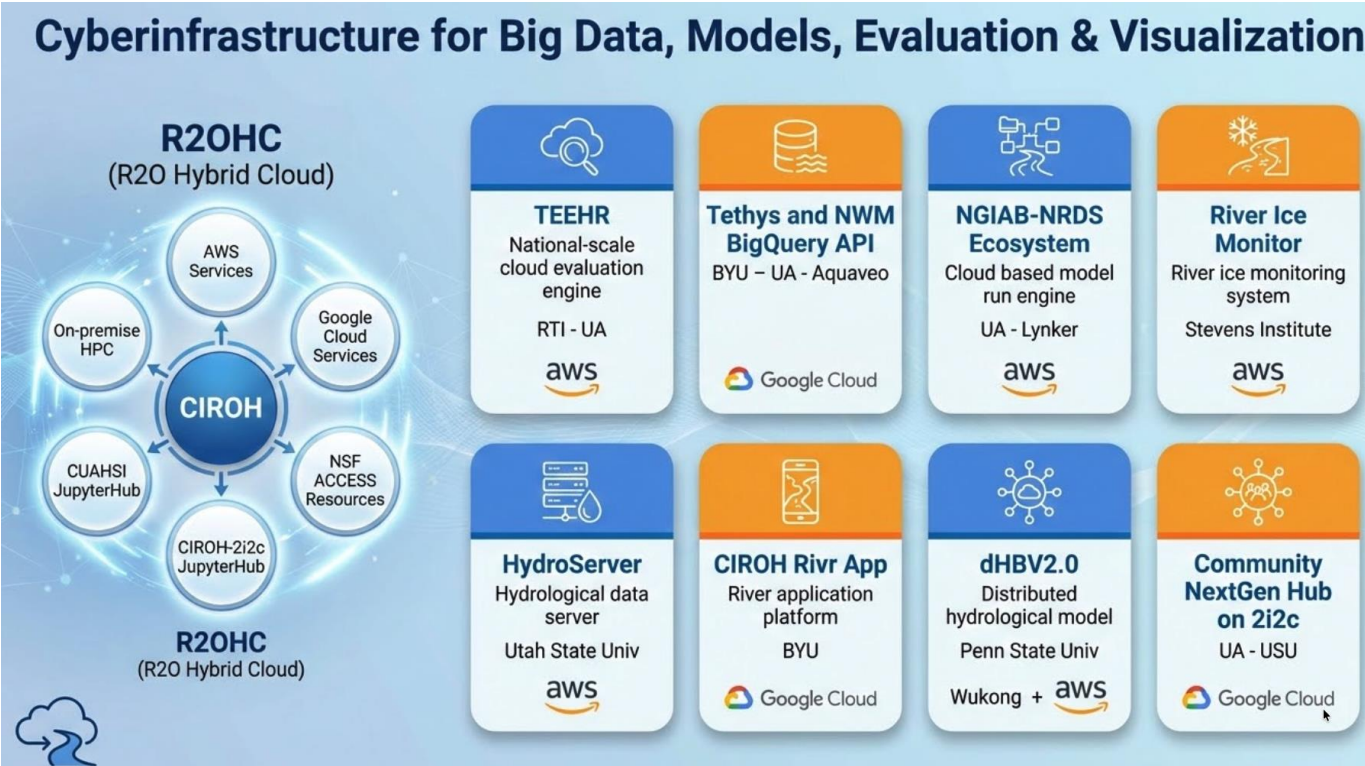
Tool stack

- GitHub Actions, Terraform, Docker, Kubernetes, AWS, GCP, Trivy, Checkov, and Gitleaks
- Standardizes DevSecOps across major CIROH

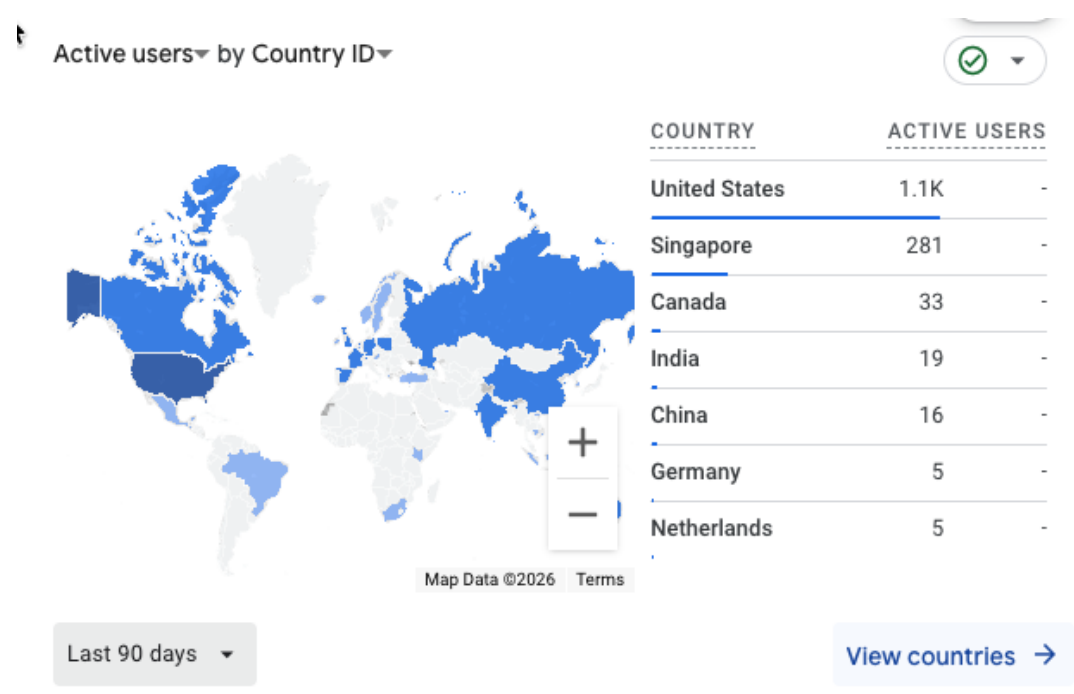


Impact At A Glance

Production research platforms are deployed across multi-cloud and HPC environments, enabling consortium-wide hydrologic modeling, evaluation, and visualization at national scale.



CIROH Hub for Access and Research Share



CIROH Hub is global – share your research and amplify your impact.

Successful beneficiary of CIROH Cyberinfrastructure



Access CIROH Cyberinfrastructure

Open to researchers, students, and partner institutions across the consortium. Scan to access onboarding documentation and submit your resource request at hub.ciroh.org/docs/services/access.



Call to Community

Current Cyberinfrastructure Users:

- We encourage you to write CIROH Hub blogs about the research supported using the cyberinfrastructure.
- Inform us when your project ends so we can archive your resources appropriately.
- Keep us updated about your research progress any way you can!



IT Support

Events supported

- CIROH DevCon 2023, 2024, 2025, and 2026
- CIROH Summer Institute 2024 and 2025
- REU 2024 and 2025

Support channels

- CIROH AWS Office Hours
- CIROH IT Slack
- Email support — ciroh-it-support@ua.edu

Resources

- CIROH Hub
- Monthly Blog and News

Partners



Acknowledgments

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- Aquaveo, CUAHSI, Lynker, UA OIT and HPC Center and participants from community resources working group have contributed key suggestions and development for this project

Research to Operations (R2O) Hybrid Cloud

CIROH Cyberinfrastructure enables:

- Cloud Resources for scalable data processing
- High-performance computing (HPC) for intensive simulations
- Storage capabilities for large-scale hydrological datasets.

